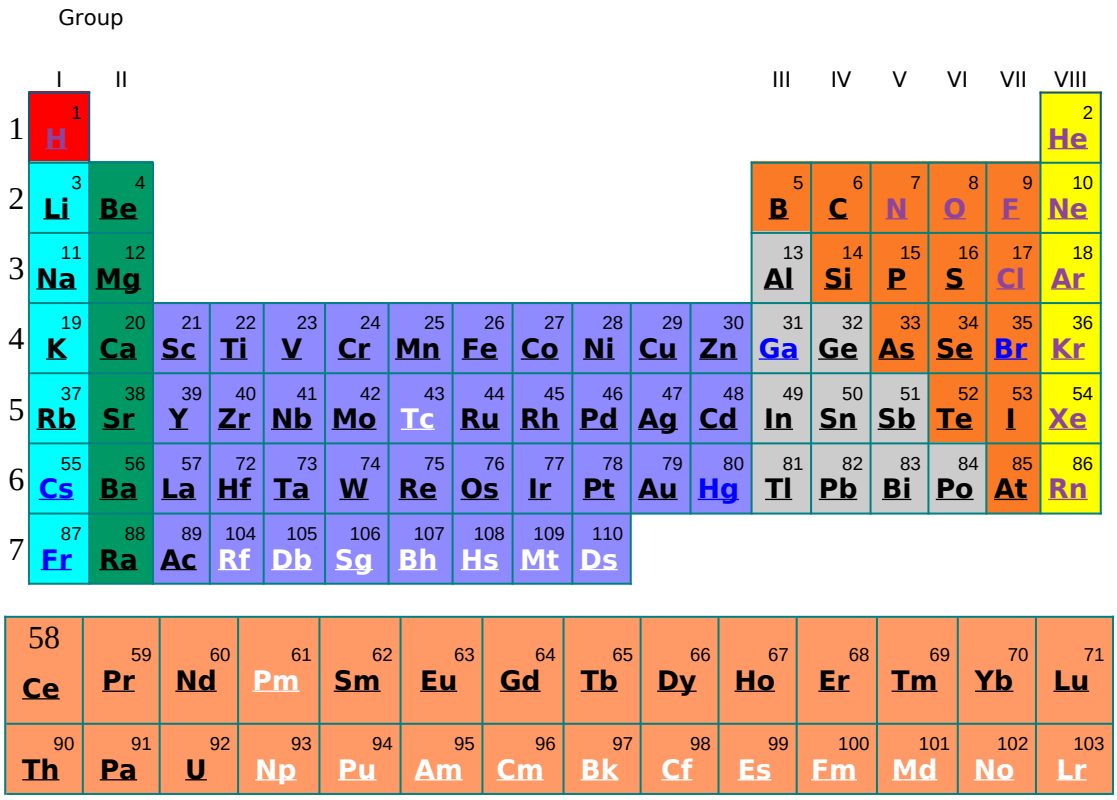


Legend

Solid Cs Liquid H Gas Tc Synthetic

Li

Alkali metals Alkali earth metals Transition metals Rare earth metals

Other metals Noble gases Halogens Other nonmetals

Periodic Table first discovered in 1869 by Dmitry I. Mendeleyev is a way of presenting all the elements so as to show their similarities and differences. The elements are arranged in increasing order of **atomic number(Z)** as you go from left to right accross the table. The horizontal rows a called **periods** and the vertical rows, **groups.**

A **noble gas** is found at the right hand side of each period. There is a progression from metals to non-metals across each period. Elements found in groups (e.g. alkali, halogens) have a similar

electronic configuration. The number of electrons in outer shell is the same as the number of the group (e.g. lithium 2·1).
The block of elements between groups II and III are called **transition metals**. These are similar in many ways; they produce colored compounds, have variable valency and are often used as **catalysts**. Elements 58 to 71 are known as **lanthanide** or rare earth elements. These elements are found on earth in only very small amounts.
Elements 90 to 103 are known as the actinide elements. They include most of the will known elements which are found in **nuclear reactions**. The elements with larger atomic numbers than 92 do not occur naturally. They have all been produced artificially by bombarding other elements with particles